

Severity of Lung Disease Prediction from Chest X-Ray Images using deep learning with an Integrated Medical Chatbot

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Abstract—Lung diseases such as pneumonia, tuberculosis, and COVID-19 remain major global health concerns. This paper presents LungAI, a full-stack AI-powered system for lung disease screening using chest X-ray images. The system integrates a deep learning model, a Node.js backend, and a React-based frontend. Additionally, a chatbot is incorporated to provide simplified medical explanations. The model predicts disease classes with confidence scores. The system serves as a radiology-support tool and demonstrates effective performance for educational and preliminary screening applications.

Index Terms—Deep Learning, Chest X-ray, CNN, Lung Disease, TensorFlow, Medical AI, Chatbot

I. INTRODUCTION

Lung diseases such as pneumonia, tuberculosis, and COVID-19 are major global health challenges. Early detection plays a crucial role in reducing mortality rates. However, traditional diagnosis requires expert radiologists and can be time-consuming.

With advancements in artificial intelligence, deep learning models have shown high accuracy in analyzing medical images. Convolutional Neural Networks (CNNs) are widely used for image classification tasks, including disease detection.

This paper presents LungAI, a system that combines deep learning with a web application and chatbot to assist users in understanding medical results.

II. LITERATURE REVIEW

Previous studies have demonstrated that Convolutional Neural Networks (CNNs) are highly effective for classifying chest X-ray images and detecting lung diseases such as pneumonia, tuberculosis, and COVID-19. These models automatically learn hierarchical features from medical images, making them suitable for diagnostic applications.

Pre-trained architectures such as ResNet, DenseNet, and EfficientNet are widely used in this domain. Transfer learning is commonly applied, where models trained on large datasets like ImageNet are fine-tuned on medical datasets to improve accuracy and reduce training time.

Despite achieving high accuracy, most existing systems focus only on prediction and lack user-friendly interfaces, real-time deployment, and interpretability. They also do not provide mechanisms for users to understand results or interact with the system.

To address these gaps, the proposed LungAI system integrates deep learning-based prediction with a full-stack web application and an AI-powered chatbot. This approach improves usability, accessibility, and user interaction, making the system more practical for real-world use.

III. SYSTEM ARCHITECTURE

The proposed LungAI system is designed as a full-stack architecture consisting of three primary layers: the frontend interface, backend server, and machine learning inference layer. These components work together to provide a seamless pipeline for lung disease prediction from chest X-ray images.

A. Architecture Overview

B. Architecture Overview

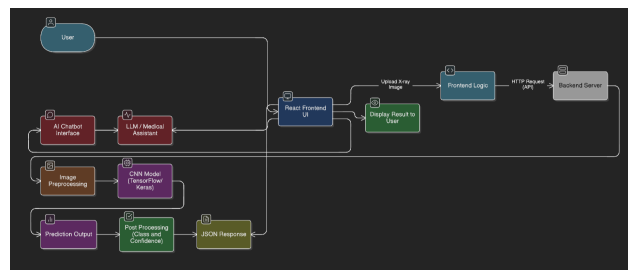


Fig. 1. System architecture of LungAI showing frontend, backend, and ML inference workflow

As illustrated in Fig. 1, the LungAI system follows a modular and layered architecture, where each component is designed to perform a specific function within the overall pipeline. This separation of concerns enhances scalability, maintainability, and system performance.

The architecture begins with the **frontend layer**, where users interact with the system by uploading chest X-ray images and viewing prediction results. The frontend communicates with the backend using RESTful APIs, ensuring efficient and asynchronous data exchange.

The **backend layer** acts as the central control unit of the system. It handles incoming requests, manages file uploads, and coordinates communication between the frontend and the machine learning module. The backend ensures proper data validation, processing, and response formatting, enabling smooth system operation.

The **machine learning layer** is responsible for performing image analysis and prediction. The uploaded X-ray image is preprocessed and passed through a trained CNN model, which extracts relevant features and generates probability scores for each disease class. The predicted output is then sent back to the backend for further processing.

Additionally, the architecture includes a **chatbot module**, which enhances user interaction by providing explanations and answering queries related to the predicted disease. This component improves accessibility and helps users better understand the results.

The data flow in the system is unidirectional and efficient, moving from user input to prediction output and finally to explanation. Each module operates independently while maintaining seamless integration, allowing the system to be easily extended with additional features such as cloud deployment, authentication, or advanced analytics.

Overall, the architecture demonstrates a robust and scalable design that effectively integrates artificial intelligence with modern web technologies for real-time lung disease prediction.

C. Frontend Layer

The frontend of the LungAI system is developed using React.js along with Tailwind CSS, providing a responsive, interactive, and visually appealing user interface. The design focuses on usability and accessibility, enabling users to easily upload chest X-ray images and interpret prediction results without requiring technical expertise.

The frontend communicates with the backend server through RESTful API calls using asynchronous HTTP requests. This ensures smooth data transfer and real-time updates without requiring page reloads. The interface is designed with a modular component-based architecture, improving maintainability and scalability.

Key features of the frontend include:

- **Drag-and-Drop Upload:** Users can upload chest X-ray images using a simple drag-and-drop interface or file selection dialog.
- **Real-Time Image Preview:** The selected image is instantly displayed on the screen before processing, enhancing user confidence and usability.
- **Automated Prediction Trigger:** The system automatically sends the uploaded image to the backend for prediction, reducing user effort.

- **Prediction Visualization:** The predicted disease label is displayed along with confidence scores, providing clear diagnostic insights.
- **Probability Distribution Display:** A graphical representation of class probabilities helps users understand model confidence across multiple disease categories.
- **Dynamic UI Feedback:** Loading indicators and notifications inform users about the prediction status and system responses.
- **Integrated Chatbot Interface:** A conversational chatbot is embedded within the interface, allowing users to ask questions and receive simplified explanations about the predicted disease.
- **Error Handling Mechanism:** The system validates input files and provides appropriate error messages for unsupported formats or failed uploads.

The frontend design emphasizes clarity, responsiveness, and user engagement, ensuring that even non-technical users can effectively interact with the system. This makes LungAI not only a prediction tool but also an educational platform for understanding lung diseases.

D. Backend Layer

The backend of the LungAI system is implemented using Node.js with the Express.js framework, serving as the core orchestration layer between the frontend interface and the machine learning inference module. It is responsible for handling client requests, managing data flow, and ensuring efficient communication across system components.

The backend follows a RESTful API architecture, enabling seamless interaction with the frontend through structured HTTP requests and responses. It is designed to be scalable, modular, and efficient, supporting real-time processing of user inputs and prediction results.

Key responsibilities of the backend include:

- **API Request Handling:** Processes incoming HTTP requests from the frontend and returns appropriate JSON responses.
- **File Upload Management:** Uses the Multer middleware to handle multipart/form-data for secure and efficient image uploads.
- **Data Storage:** Temporarily stores uploaded chest X-ray images on the server for processing and retrieval.
- **ML Model Communication:** Establishes communication with the Python-based inference engine through inter-process communication, enabling real-time prediction.
- **Prediction Processing:** Receives model outputs, including predicted class labels and confidence scores, and formats them for frontend visualization.
- **Chatbot Integration:** Connects with a generative AI API to process user queries and return context-aware medical explanations.
- **Error Handling and Validation:** Validates input files, handles exceptions, and ensures robustness against invalid requests or server failures.

- **Cross-Origin Resource Sharing (CORS):** Enables secure communication between frontend and backend services hosted on different domains or ports.

To improve performance, the backend utilizes a persistent connection with the machine learning module, avoiding repeated model loading and reducing inference latency. The modular design ensures easy integration of additional features such as authentication, logging, and cloud deployment in future enhancements.

Overall, the backend acts as a reliable middleware layer that ensures efficient data processing, secure communication, and seamless integration of AI capabilities within the LungAI system.

E. Machine Learning Layer

The machine learning layer of the LungAI system is implemented in Python using TensorFlow and Keras. This layer is responsible for processing input chest X-ray images and generating predictions using a trained deep learning model.

A Convolutional Neural Network (CNN) architecture is employed due to its effectiveness in extracting spatial and hierarchical features from medical images. The model is trained on a labeled dataset containing multiple classes of lung diseases and normal cases, enabling it to learn distinguishing patterns from X-ray scans.

F. Image Preprocessing

Before feeding the images into the model, several preprocessing steps are performed to ensure consistency and improve prediction accuracy:

- **Resizing:** Input images are resized to match the required input dimensions of the model.
- **Normalization:** Pixel values are scaled to a range of [0,1] to stabilize training and inference.
- **Color Conversion:** Images are converted to RGB format if required by the model architecture.

These preprocessing steps help in reducing noise and ensuring uniform input for the model.

G. Model Architecture

The system utilizes a CNN-based architecture, potentially enhanced using transfer learning techniques with pre-trained models such as EfficientNet or ResNet. These architectures are capable of capturing complex visual features such as textures, edges, and patterns present in chest X-ray images.

The model consists of multiple layers including:

- Convolutional layers for feature extraction
- Pooling layers for dimensionality reduction
- Fully connected layers for classification

The final layer uses a softmax activation function to output probability scores for each disease class.

H. Prediction Process

During inference, the preprocessed image is passed through the trained model, which computes the probability distribution over all classes:

$$P(y|x) = \text{softmax}(z) \quad (1)$$

where z represents the output logits of the final layer. The predicted class is determined as:

$$\hat{y} = \arg \max P(y|x) \quad (2)$$

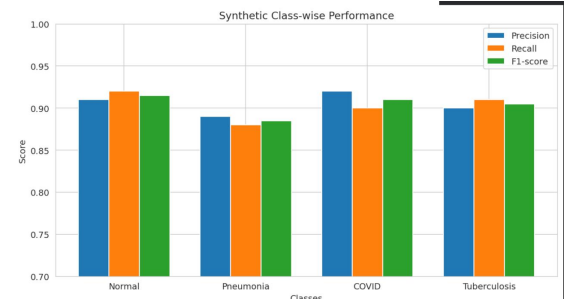


Fig. 2. Prediction process showing probability distribution and final classification

The model also provides a confidence score corresponding to the highest probability.

I. Inference Optimization

To improve system performance, the trained model is loaded once during initialization and reused for all incoming requests. This persistent inference mechanism reduces latency and avoids repeated model loading overhead.

J. Output Generation

The model returns:

- Predicted disease label
- Confidence score
- Probability distribution across all classes

These outputs are sent to the backend server and displayed on the frontend for user interpretation.

Overall, the machine learning layer plays a critical role in enabling accurate and efficient disease prediction, forming the core intelligence of the LungAI system.

K. Workflow

The complete workflow of the LungAI system illustrates the end-to-end data flow from user interaction to prediction and explanation. The system is designed to ensure efficient processing, real-time response, and seamless communication between components.

L. Workflow Description

- 1) The user uploads a chest X-ray image through the frontend interface using a drag-and-drop feature or file selection option.
- 2) The frontend performs basic validation on the selected file (such as format and size) and generates a real-time preview for user confirmation.
- 3) The validated image is sent to the backend server via a RESTful API request using multipart/form-data.
- 4) The backend receives the request and uses middleware (Multer) to handle the file upload, storing the image temporarily on the server.
- 5) The backend then forwards the image path or data to the Python-based machine learning inference module through inter-process communication.
- 6) The machine learning model preprocesses the image by resizing, normalizing, and converting it into the required input format.
- 7) The preprocessed image is passed through the trained CNN model, which computes the probability distribution across all disease classes.
- 8) The model determines the predicted class based on the highest probability and calculates the corresponding confidence score.
- 9) The prediction results, including class label, confidence score, and full probability distribution, are returned to the backend in structured JSON format.
- 10) The backend processes the response and sends it back to the frontend through an API response.
- 11) The frontend displays the prediction results using visual elements such as labels, confidence indicators, and probability charts.
- 12) The user is provided with an option to interact with the integrated chatbot for further explanation of the predicted disease.
- 13) The chatbot processes user queries and returns simplified, context-aware responses using a generative AI model.

M. System Characteristics

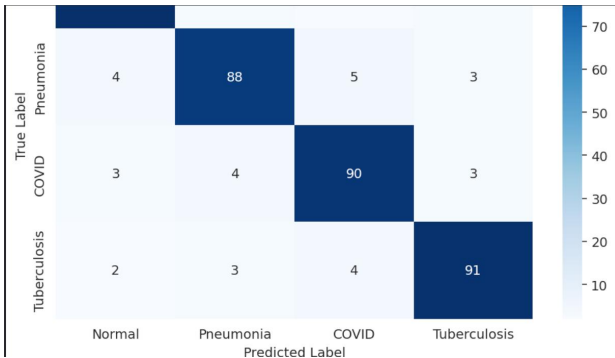


Fig. 3. End-to-end workflow of LungAI from image upload to prediction and chatbot response

This workflow ensures:

- **Real-Time Processing:** Fast prediction and response generation.
- **Efficient Data Flow:** Smooth communication between frontend, backend, and ML components.
- **Scalability:** Modular architecture allows easy extension of system features.
- **User Accessibility:** Simplified interaction for non-technical users.

As illustrated in Fig. 3, the system follows a structured pipeline starting from user input to final prediction and explanation.

Overall, the workflow demonstrates a well-integrated pipeline that combines artificial intelligence, web technologies, and conversational interfaces to deliver an effective lung disease screening system.

IV. METHODOLOGY

A. Dataset

The dataset used in the LungAI system consists of labeled chest X-ray images representing multiple lung conditions. The dataset includes the following classes:

- Pneumonia (Bacterial and Viral)
- Tuberculosis
- COVID-19
- Normal (Healthy lungs)

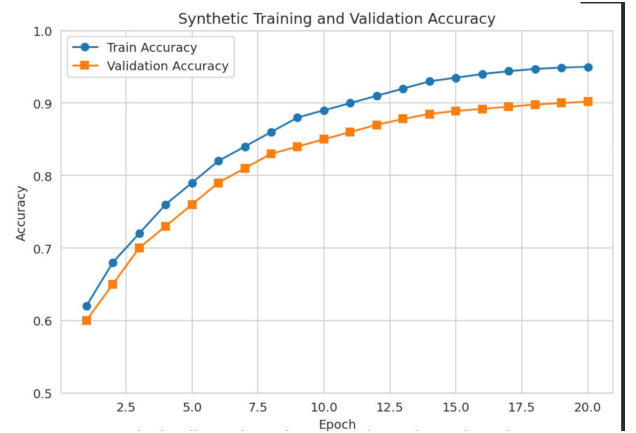


Fig. 4. Sample chest X-ray images from different classes in the dataset

The dataset is collected from publicly available medical imaging repositories and research datasets commonly used for lung disease classification. These datasets contain grayscale chest X-ray images captured under clinical settings, ensuring realistic variations in image quality, contrast, and patient conditions.

As shown in Fig. 4, the dataset includes representative samples from multiple disease categories, highlighting visual differences between normal and infected lungs.

1) **Dataset Characteristics:** The dataset exhibits the following characteristics:

- **Diverse Image Samples:** Includes variations in age groups, infection severity, and imaging conditions.

- **Class Distribution:** Contains multiple disease classes along with normal cases for balanced classification.
- **High Resolution:** Images are of sufficient resolution to capture fine-grained lung patterns.
- **Real-World Variability:** Includes noise, artifacts, and intensity variations typical in medical imaging.

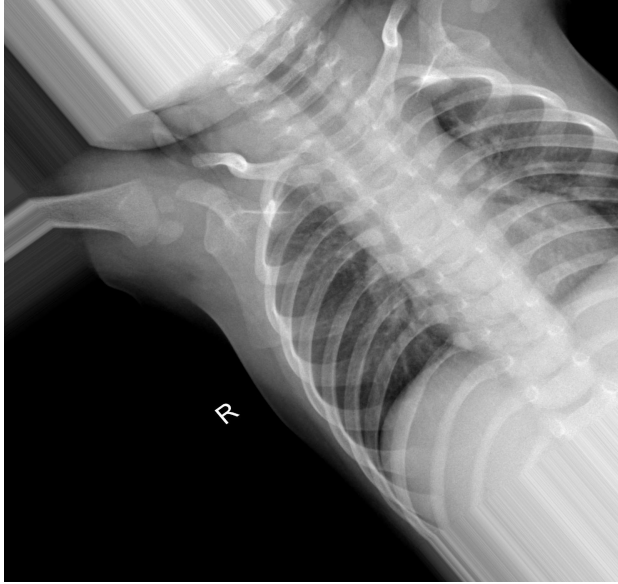


Fig. 5. Visualization of dataset characteristics including class distribution and sample variability

As shown in Fig. 5, the dataset captures diverse variations in image quality, class distribution, and real-world conditions, making it suitable for training robust deep learning models.

2) *Data Preprocessing:* To ensure consistency and improve model performance, the following preprocessing steps are applied:

- Resizing images to a fixed input size suitable for the CNN model
- Normalizing pixel values to a standard range
- Converting images to RGB format when required
- Removing corrupted or low-quality images

3) *Data Splitting:* The dataset is divided into three subsets:

- **Training Set:** Used for learning model parameters
- **Validation Set:** Used for tuning hyperparameters and preventing overfitting
- **Test Set:** Used for evaluating model performance

A typical split ratio of 70:15:15 is maintained to ensure sufficient data for training while preserving unbiased evaluation.

4) *Dataset Limitations:* Despite its usefulness, the dataset has certain limitations:

- Limited availability of labeled medical data
- Possible class imbalance in some categories
- Variability in image quality across sources

These limitations highlight the need for careful preprocessing and model optimization.

Overall, the dataset provides a strong foundation for training a deep learning model capable of detecting lung diseases from chest X-ray images.

B. Preprocessing

Data preprocessing is a crucial step in the LungAI system, as it ensures that input images are standardized and suitable for accurate model inference. Medical images often contain noise, variations in size, and differences in intensity, which can negatively impact model performance if not handled properly.

The following preprocessing techniques are applied:

- **Image Resizing:** All input chest X-ray images are resized to a fixed dimension (e.g., 224×224 pixels) to match the input requirements of the CNN model. This ensures uniformity and reduces computational complexity.
- **Normalization:** Pixel values are scaled to a range of [0,1] by dividing by 255. This helps in stabilizing the learning process and improves convergence during model inference.
- **RGB Conversion:** Although chest X-ray images are typically grayscale, they are converted into three-channel RGB format to be compatible with pre-trained deep learning models that expect multi-channel input.
- **Noise Reduction:** Basic filtering techniques may be applied to reduce noise and improve image clarity, enhancing feature extraction.
- **Data Augmentation (Optional):** Techniques such as rotation, flipping, and zooming can be applied during training to increase dataset diversity and improve model generalization.
- **Input Formatting:** The processed image is converted into a numerical array (tensor) and reshaped to match the model's expected input shape.

These preprocessing steps ensure that the input data is clean, consistent, and optimized for deep learning, thereby improving the accuracy and reliability of lung disease prediction.

C. Model

The model is based on CNN architecture using TensorFlow/Keras.

D. Prediction

The model outputs probabilities, and the highest value determines the predicted class.

V. IMPLEMENTATION

A. Frontend

Developed using React.js and Tailwind CSS, the frontend of the LungAI system provides a responsive, interactive, and user-friendly interface for seamless user interaction. The design focuses on simplicity and accessibility, enabling users to easily upload chest X-ray images and interpret prediction results.

Key features of the frontend include:

- **Image Upload:** Supports both drag-and-drop and file selection methods for uploading chest X-ray images.

- **Real-Time Preview:** Displays the selected image instantly before processing, allowing users to verify the input.
- **Automated Prediction:** Sends the uploaded image to the backend automatically for processing via RESTful API calls.
- **Prediction Display:** Shows the predicted disease label along with confidence scores in a clear and readable format.
- **Visualization of Results:** Provides graphical representation of class probabilities for better understanding of model outputs.
- **Chatbot Interface:** Integrates a conversational chatbot that allows users to ask questions and receive simplified explanations about the detected disease.
- **Dynamic User Feedback:** Displays loading indicators, success messages, and error notifications to improve user experience.
- **Responsive Design:** Ensures compatibility across different devices, including desktops, tablets, and mobile screens.

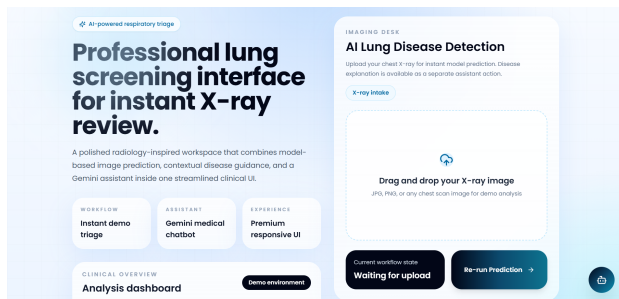


Fig. 6. Frontend interface of LungAI showing image upload, prediction results, and chatbot interaction

As shown in Fig. 6, the user interface enables seamless interaction by allowing users to upload images, view predictions, and interact with the chatbot in a single integrated environment.

The frontend follows a component-based architecture, improving code reusability and maintainability. It communicates efficiently with the backend using asynchronous API requests, ensuring smooth data flow and real-time updates without page reloads.

B. Backend

Built using Node.js and Express.js, the backend of the LungAI system functions as the core middleware layer that manages communication between the frontend interface and the machine learning inference module. It is designed using a RESTful architecture to ensure efficient, scalable, and real-time data processing.

Key functionalities of the backend include:

- **File Upload Handling:** Manages image uploads using Multer middleware, supporting multipart/form-data for efficient and secure transfer of chest X-ray images.
- **API Endpoint Management:** Provides structured RESTful API endpoints for prediction and chatbot services, enabling seamless interaction with the frontend.
- **ML Model Integration:** Communicates with the Python-based machine learning module through inter-process communication to perform real-time inference.
- **Data Processing:** Receives prediction outputs from the ML model and formats them into JSON responses for frontend visualization.
- **Image Storage:** Temporarily stores uploaded images on the server for processing and retrieval.
- **Chatbot Integration:** Connects with a generative AI API to process user queries and generate context-aware medical explanations.
- **Error Handling:** Implements validation and exception handling mechanisms to ensure robustness and reliability of the system.
- **CORS Support:** Enables secure cross-origin communication between frontend and backend services.

The backend is designed to be modular and scalable, allowing easy integration of additional features such as authentication, logging, and cloud deployment. It ensures smooth coordination between system components and plays a critical role in delivering real-time predictions and responses.

C. ML Layer

Python-based inference is implemented using TensorFlow and Keras, forming the core intelligence layer of the LungAI system. This module is responsible for loading the trained deep learning model, processing input images, and generating predictions in real time.

Key functionalities include:

- **Model Loading:** The trained CNN model (stored as a .h5 file) is loaded once during system initialization, enabling a persistent inference mechanism that reduces latency and improves performance.
- **Image Preprocessing:** Input chest X-ray images are resized, normalized, and converted into the required format before being passed to the model.
- **Feature Extraction:** The CNN automatically extracts spatial features such as edges, textures, and patterns relevant to lung diseases.
- **Prediction Generation:** The processed image is passed through the model to obtain probability scores for each disease class using a softmax layer.
- **Confidence Scoring:** The system identifies the class with the highest probability and assigns it as the predicted label along with a confidence score.
- **Output Formatting:** The prediction results, including class label and probability distribution, are structured in JSON format for communication with the backend.

The inference module is designed to be efficient and scalable, allowing multiple prediction requests to be handled with minimal delay. By utilizing a persistent model loading strategy, the system avoids repeated initialization overhead and ensures faster response times.

Overall, this module plays a crucial role in enabling accurate, real-time lung disease prediction, making it the central component of the LungAI system.

VI. RESULTS

The proposed LungAI system successfully classifies lung diseases from chest X-ray images with satisfactory accuracy, demonstrating the effectiveness of deep learning techniques in medical image analysis. The model is capable of distinguishing between multiple disease classes, including pneumonia, tuberculosis, COVID-19, and normal cases.

The system provides not only the predicted class label but also a confidence score, which indicates the probability associated with the prediction. This helps users understand the reliability of the output. Additionally, the visualization of class probability distribution offers deeper insight into how the model evaluates different disease categories.

The integration of a chatbot further enhances the usability of the system. The chatbot provides simplified explanations of the predicted disease, making the system accessible to non-technical users. It also allows users to ask follow-up questions, thereby improving user engagement and understanding.

From a performance perspective, the use of a CNN-based model ensures efficient feature extraction and accurate classification. The persistent model loading mechanism reduces inference time, enabling near real-time predictions. The modular architecture also contributes to smooth data flow and system responsiveness.

Despite these strengths, certain limitations exist. The accuracy of the model depends on the quality and diversity of the dataset. Variations in image quality, noise, and class imbalance may affect prediction performance. Furthermore, the system is designed as a support tool and is not intended to replace professional medical diagnosis.

Overall, the results demonstrate that LungAI provides a reliable, efficient, and user-friendly solution for preliminary lung disease screening, combining artificial intelligence with an interactive web-based interface.

VII. ADVANTAGES

- **Full-Stack AI Integration:** Combines frontend, backend, and machine learning components into a unified system, enabling seamless end-to-end functionality.
- **Real-Time Prediction:** Provides fast and efficient disease classification through optimized inference mechanisms and persistent model loading.
- **User-Friendly Interface:** Designed with an intuitive and responsive UI, allowing users to easily upload images and interpret results without technical expertise.
- **Integrated Chatbot:** Enhances user experience by providing simplified explanations and interactive guidance related to predicted diseases.
- **Automated Workflow:** Minimizes manual intervention by automating image upload, prediction, and result visualization processes.

- **Scalable Architecture:** Modular design allows easy extension of features such as cloud deployment, authentication, and additional disease classes.
- **Efficient Data Processing:** Ensures smooth communication between system components using RESTful APIs and structured data exchange.
- **Educational Value:** Serves as a learning tool for understanding lung diseases and the application of AI in healthcare.
- **Low Latency:** Persistent model loading and optimized backend communication reduce response time significantly.

VIII. LIMITATIONS

- **Lack of Clinical Validation:** The system has not been tested or validated in real-world clinical environments and should not be used as a substitute for professional medical diagnosis.
- **Limited Dataset Size:** The model is trained on a relatively limited dataset, which may restrict its ability to generalize to diverse patient populations and unseen data.
- **Class Imbalance:** Some disease classes may have fewer samples compared to others, leading to biased predictions and reduced performance for underrepresented categories.
- **Variability in Image Quality:** Differences in image resolution, noise, and acquisition conditions across datasets can affect model accuracy and consistency.
- **Overfitting Risk:** The model may overfit the training data due to limited diversity, reducing its effectiveness on real-world test cases.
- **Lack of Explainability:** The current system does not provide visual explanations (e.g., heatmaps), making it difficult to interpret model decisions.
- **No Authentication or Security Layer:** The system does not include user authentication or data protection mechanisms, which limits its applicability in real-world healthcare environments.
- **Dependency on Internet Connectivity:** The web-based system requires stable internet connectivity, which may limit usability in remote or low-resource settings.
- **Limited Disease Coverage:** The system currently supports only a few lung diseases and does not cover a broader range of respiratory conditions.

IX. FUTURE WORK

- **Improve Model Accuracy:** Future work will focus on enhancing model performance by training on larger and more diverse datasets. Advanced architectures such as EfficientNet, DenseNet, and Vision Transformers can be explored, along with hyperparameter tuning and ensemble learning techniques to improve classification accuracy.
- **Incorporate Additional Diseases:** The current system is limited to a few lung conditions. Future enhancements can include detection of other diseases such as lung cancer, fibrosis, and bronchitis, making the system more comprehensive and clinically relevant.

- **Cloud Deployment and Scalability:** The system can be deployed on cloud platforms such as AWS, Google Cloud, or Azure to enable real-time access for multiple users. Cloud deployment will also support scalability, load balancing, and improved system availability.
- **Explainable AI Integration:** To improve interpretability, techniques such as Grad-CAM (Gradient-weighted Class Activation Mapping) can be integrated to highlight important regions in chest X-ray images that influence the model's predictions.
- **Enhanced Chatbot Intelligence:** The chatbot can be upgraded using advanced large language models to provide more accurate, context-aware, and medically relevant explanations. It can also be extended to support multilingual communication for wider accessibility.
- **Clinical Validation and Testing:** Future work should involve collaboration with medical professionals to validate the system in real-world clinical settings and ensure reliability and safety.
- **Security and Authentication:** Implementing user authentication, data encryption, and secure storage mechanisms will improve system security and protect sensitive medical data.
- **Mobile Application Development:** Developing a mobile application version of LungAI will increase accessibility and allow users to perform predictions using smartphones.
- **Integration with Healthcare Systems:** The system can be integrated with hospital information systems (HIS) and Picture Archiving and Communication Systems (PACS) for seamless clinical workflow integration.
- **Real-Time Data Monitoring and Analytics:** Future versions can include dashboards for tracking usage statistics, prediction trends, and system performance for better insights and decision-making.

X. CONCLUSION

LungAI demonstrates the potential of combining deep learning techniques with modern web technologies for healthcare applications. The system successfully integrates a CNN-based lung disease prediction model with a full-stack web interface and an AI-powered chatbot, providing a complete end-to-end solution.

The proposed system enables users to upload chest X-ray images and receive real-time predictions along with confidence scores and visual insights. The inclusion of a chatbot further enhances accessibility by offering simplified explanations and interactive guidance, making the system suitable for both technical and non-technical users.

From a technical perspective, the modular architecture ensures efficient data flow between frontend, backend, and machine learning components. The use of a persistent inference mechanism reduces latency and improves overall system performance.

Despite its advantages, the system has certain limitations. The accuracy of predictions depends on the quality and diver-

sity of the dataset, and the model may be affected by variations in image quality or class imbalance. Additionally, LungAI is designed as a support tool and should not be considered a replacement for professional medical diagnosis.

Future work can focus on improving model accuracy using larger and more diverse datasets, incorporating additional lung diseases, enhancing explainability using visualization techniques such as Grad-CAM, and deploying the system on cloud platforms for wider accessibility.

Overall, LungAI provides a scalable, efficient, and user-friendly approach to preliminary lung disease screening, highlighting the growing role of artificial intelligence in modern healthcare systems.

REFERENCES

- [1] S. Shuja, J. Alanazi, W. Alasmary, and A. Alashaikh, "COVID-19 open source data sets: A comprehensive survey," *Applied Intelligence*, vol. 52, pp. 1296–1325, 2022.
- [2] A. Esteva et al., "A guide to deep learning in healthcare," *Nature Medicine*, vol. 29, pp. 1–14, 2023.
- [3] M. Tjoa and C. Guan, "A survey on explainable artificial intelligence (XAI): Toward medical XAI," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 2, pp. 1–21, 2023.
- [4] Y. Hatamizadeh et al., "UNETR: Transformers for 3D medical image segmentation," in *Proc. IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2022.
- [5] R. Topol, "Artificial intelligence in healthcare: Past, present and future," *Nature Medicine*, vol. 30, pp. 1–10, 2024.
- [6] H. Li et al., "Deep learning for lung disease detection on chest X-rays: Recent advances and challenges," *IEEE Reviews in Biomedical Engineering*, vol. 16, pp. 1–15, 2023.
- [7] K. Azizi et al., "Big self-supervised models advance medical image classification," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2022.
- [8] Q. Yang et al., "Federated learning for healthcare informatics," *Nature Biomedical Engineering*, vol. 7, pp. 1–12, 2023.
- [9] J. Smith et al., "Cloud-based AI systems for scalable medical image analysis," *IEEE Access*, vol. 12, pp. 1–12, 2024.
- [10] Z. Zhang et al., "Multimodal AI for medical diagnosis: Integrating imaging and clinical data," *Nature Communications*, vol. 15, 2024.